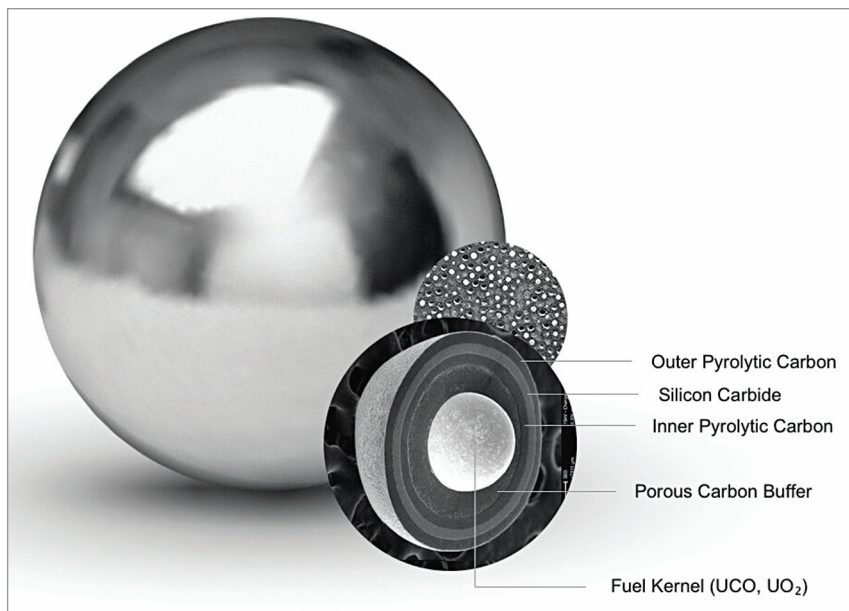


using technologies such as high-temperature gas-cooled reactors, molten salt reactors, and compact fast-spectrum reactors. These are intended for captive applications, including industrial process heat, off-grid electricity, and hydrogen production.

The Bhabha Atomic Research Centre is developing a 220MWe pressurised water reactor based on the proven IPHWR-220 technology. The Nuclear Power Corporation of India Limited is preparing the detailed project report for BSMR deployment. The Atomic Energy Regulatory Board is reviewing its approval and licensing procedures to accommodate SMR-specific designs and multi-module operations.

Internationally, India is actively collaborating with other countries to complement its indigenous capabilities. Joint technical working groups have been established with the United States for cooperation on light-water small modular reactors (SMRs) and non-light-water advanced reactor designs. Negotiations are underway with France to explore high-temperature reactor technologies, particularly for industrial use. Discussions are also ongoing with Russia concerning compact fast-spectrum reactors. India is working with the International Atomic Energy Agency (IAEA) to help develop global safety and



Advanced SMR designs require new fuel types, such as Tristructural-Isotropic (TRISO)
(Source: <https://x-energy.com>)

licensing frameworks.

Through coordinated efforts in domestic innovation, regulatory modernisation, policy reform, and international cooperation, India is positioning itself as a leading global centre for SMR development and deployment.

Challenges in unlocking the potential of SMRs

While SMRs promise several advantages, there are challenges and barriers—ranging from regulatory hurdles to technological integration and financing constraints—that make widespread adoption difficult.

One of the foremost hurdles is the approval, licensing, and regulatory processes, which are site-specific and originally drafted for large nuclear plants. These processes need to be reviewed to make them technology-neutral, design-adaptive, and supportive of modular deployment.

The paucity of data on the actual performance of SMRs makes it challenging to frame

regulatory processes. To address this, the IAEA is developing a more adaptive and globally aligned regulatory framework, supporting pilot projects and fostering collaboration among regulators. Nevertheless, countries face difficulties in accepting international regulations due to misalignment with their nuclear policies, liability laws and risk assessment processes. Supply chain constraints are another barrier, as critical components—such as reactor pressure vessels and advanced heat exchangers—are difficult to procure, with sources limited and concentrated in a few countries. Developing indigenous supply chains requires significant time and investment.

Additionally, the upfront costs for research, design, infrastructure, manufacturing, technology acquisition, and skilled labour are very high. Risks such as unproven designs, lack of confidence in the technology, uncertain market demand, ambiguity in the approval and licensing process, and long timelines from design to deployment contribute to financial uncertainty.

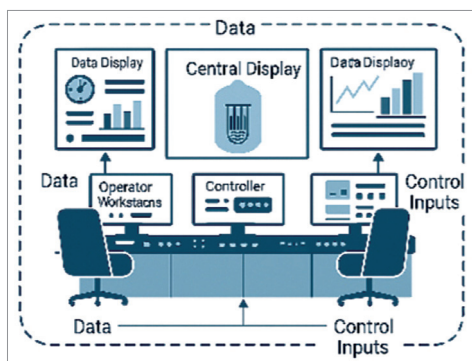


Fig. 3: SMR control room human-machine interface